REPLACING A DISTINGUISHED TRIANGLE BY AN ISOMORPHIC ONE WITH TERMWISE SPLIT ROWS

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Editorial note. Counted span is the author's own proof environment, source lines 3685-3702, following the statement at 3675-3683.

Editorial note. No omissions: both steps are displayed.

Allowed standard prerequisites (editorial).

- Cone triangle. lemma-the-same-up-to-isomorphisms (carded L26170, cited at ~3691). Provides the distinguished triangle on the cone of c.
- Third arrow of a triangle morphism. lemma-third-isomorphism-triangle (same chapter, cited at ~3698). The (id, beta, id) morphism of triangles is an isomorphism.

Source reference. Original lemma 014L: sources/stacks-derived/pinned-original-source.tex, source lines 3213–3213. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014A: sources/stacks-derived/pinned-original-source.tex, source lines 631–631. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 014S: sources/stacks-derived/pinned-original-source.tex, source lines 3537–3537. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 014D: sources/stacks-derived/pinned-original-source.tex, source lines 2675–2675. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an additive category. Let $(A^\bullet, B^\bullet, C^\bullet, a, b, c)$ be a distinguished triangle in $K(\mathcal{A})$. Then there exists an isomorphic distinguished triangle $(A^\bullet, (B')^\bullet, C^\bullet, a', b', c)$ such that $0 \to A^n \to (B')^n \to C^n \to 0$ is a split short exact sequence for all n.*

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We will use that $K(\mathcal{A})$ is a triangulated category by Proposition 014S. Let $W^\bullet$ be the cone on $c : C^\bullet \to A^\bullet[1]$ with its maps $i : A^\bullet[1] \to W^\bullet$ and $p : W^\bullet \to C^\bullet[1]$. Then $(C^\bullet, A^\bullet[1], W^\bullet, c, i, -p)$ is a distinguished triangle by Lemma 014L. Rotating backwards twice we see that $(A^\bullet, W^\bullet[-1], C^\bullet, -i[-1], p[-1], c)$ is a distinguished triangle. By TR3 there is a morphism of distinguished triangles $(\mathrm{id}, \beta, \mathrm{id}) : (A^\bullet, B^\bullet, C^\bullet, a, b, c) \to (A^\bullet, W^\bullet[-1], C^\bullet, -i[-1], p[-1], c)$ which must be an isomorphism by Lemma 014A. This finishes the proof because $0 \to A^\bullet \to W^\bullet[-1] \to C^\bullet \to 0$ is a termwise split short exact sequence of complexes by the very construction of cones in Section 014D. $\square$